

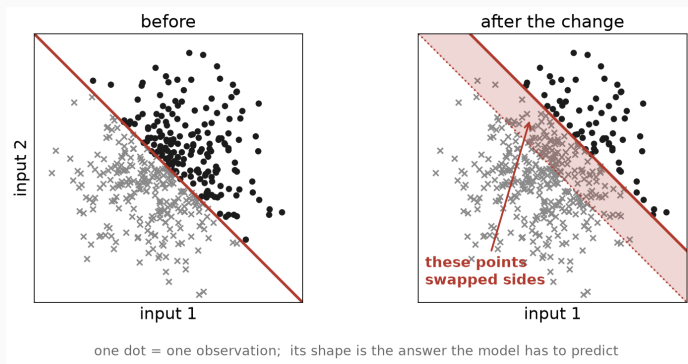
The Blind Spot Paradox

When a self-repairing model destroys the evidence its own monitor needs

Alexandre Boyer Mélaine Gouillou Salomé Fonvielle Ulysse Petit-Tichanné

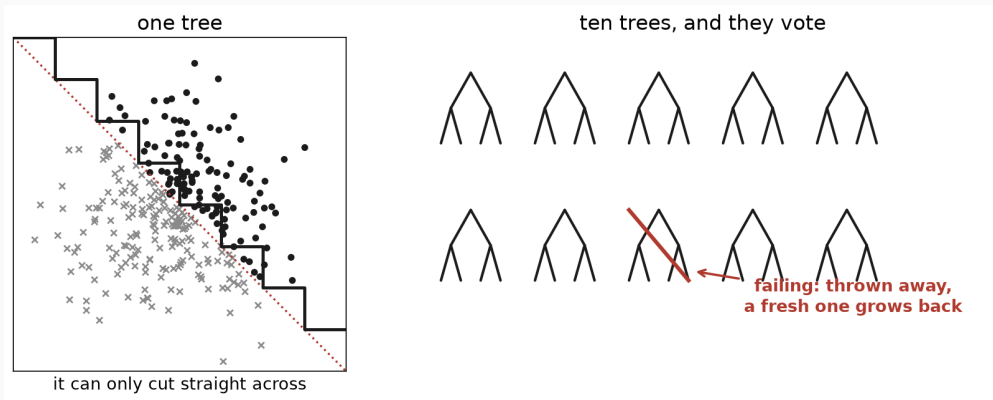
Research Track, Topic 39, CentraleSupélec. Supervisor: Raphaël Minato

Concept drift: the rule changes, the data does not



A model trained on the left keeps predicting, confidently, and is now **wrong on the shaded band**.

Defence one: a model that repairs itself

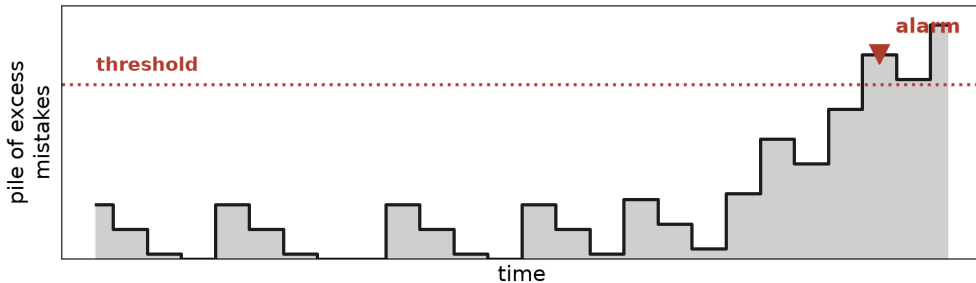


This is the **Adaptive Random Forest**. Ten trees, replaced one at a time.

Defence two: a watchdog that raises the alarm

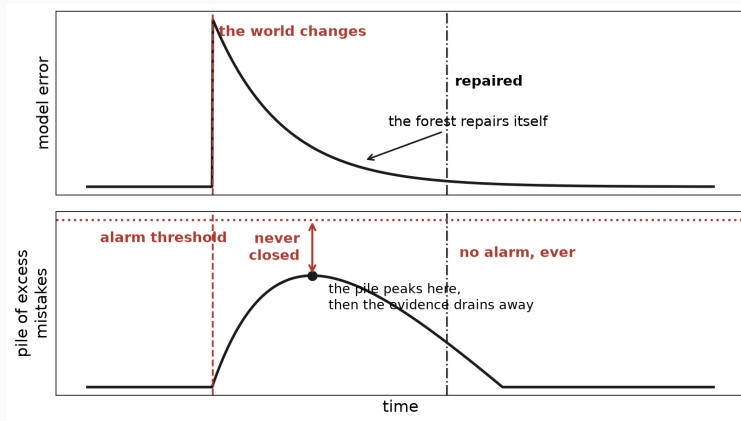
the model answers, one observation at a time (x = wrong)

x o o o x o o o o x o o o x o o x x o x x o x



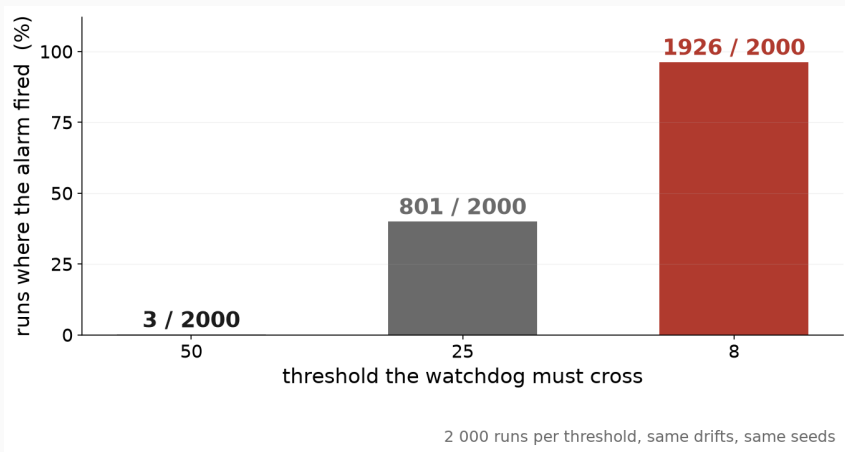
Two communities, two tools, never measured against each other.

Put them together, and they cancel out



The repair has a date. The alarm never gets one. (schematic)

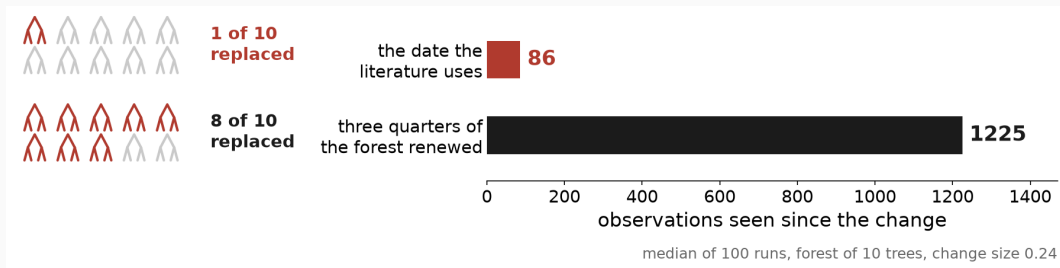
Why not just lower the threshold?



A low threshold also fires when **nothing has happened**, and an alarm that cries wolf gets switched off.

Our contribution: when is the forest actually repaired?

The alarm is what we *want*. This date measures what **stops it** from ever firing.



It fires at the *first* tree. It times the start of a repair, not its end.

Where we stand

- Yes** the alarm really does stay silent, and we measured how often. Turning the threshold down does not rescue it, it only trades silence for false alarms.
- But** the stopwatch the field uses starts at the **first tree out of ten**. It makes the forest look faster than it is, so every published figure about this race is optimistic.
- Next** put the model and the watchdog on one shared clock, and say which setting a practitioner should actually pick.
- Open** what should replace that stopwatch is a research question of its own. We have started testing candidates, and none has passed our bar yet.

Thank you.